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Production of Neutron-Rich Λ-Hypernuclei with (π^-, K^+)Reaction through Σ-Hypernuclear Formation

Theingi[1,a)], Khin Swe Myint[2,b)] and Toru Harada[3, 4,c)]

[1]*Physics Department, Mandalay University of Distance Education, Myanmar.*
[2]*Physics Department, University of Mandalay, Myanmar.*
[3]*Research Center for Physics and Mathematics, Osaka Electro-Communication University, Neyagawa, Osaka, 572-8530, Japan.*
[4]*J-PARC Branch, KEK Theory Center, Institute of Particle and Nuclear Studies, High Energy Accelerator Research Organization (KEK), 203-1, Shirakata, Tokai, Ibaraki, 319-1106, Japan.*

[a)]Corresponding author: theingimandalar@gmail.com
[b)] profkhinswemyint@gmail.com
[c)] harada@osakac.ac.jp

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Abstract. We have investigated theoretically the missing-mass spectral function of (π^-, K^+) reaction on ^{9}Be target to populate $_{\Sigma^-}{}^9$Li Σ-hypernucleus. If there is an appreciable amount of ΣN-ΛN coupling effect, this double charge exchange reaction would be a plausible way to populate neutron-rich Λ-hypernuclei. We have used Plane Wave Impulse Approximation Method together with Σ-nucleus Woods-Saxon potential. It is found that the strengths of V_Σ=-20 MeV and W_Σ=-3MeV give an observable narrow peak in the spectrum of $_{\Sigma^-}{}^9$Li. To investigate the Λ-Σ coupling effect and Σ mixing component in $_\Lambda{}^9$He, we also studied the structure calculation of $_\Lambda{}^9$He by solving the coupled channel Schrodinger equation. Λ-Σ coherent coupling effect is 0.19 MeV and Σ mixing probability is 0.2% in $_\Lambda{}^9$He.

INTRODUCTION

We consider the production of Σ-hypernuclear states in the double-charge exchange (π^-, K+) reaction on a nuclear target by the one-step mechanism. Figure 1 illustrates diagrams for the nuclear (π^-, K+) reaction by $\pi^- p \rightarrow K^+ \Sigma^-$ processes for the Σ hypernuclear states, and for Λ hypernuclear excited states. To calculate the nuclear (π^-, K+) missing-mass spectra within a plane wave impulse approximation, we employ the Green's function method which is one of the most powerful treatments in the calculation of a spectrum in which not only bound states but also continuum states are described with an attractive potential for spreading components.

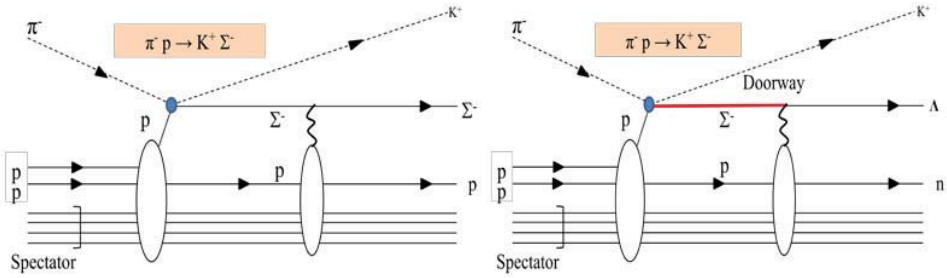

FIGURE 1. Diagrams for double-charge exchange (π^-, K^+) reactions on nuclear targets, leading to production of (a) Σ hypernuclear states by $\pi^- p \rightarrow K^+ \Sigma^-$ processes and (b). Λ hypernuclear states by $\Sigma^- p \leftrightarrow \Lambda n$ processes.

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COMPUTATIONAL PROCEDURES

Missing-Mass Spectrum of (π^-, K^+) Reaction

The transition matrix between the initial state $\pi^- + {}^9$Be and the final state $K^+ + {}_{\Sigma}{}^{9}He(\Sigma^- - {}^8\text{Li})$ is written as

$$T_{fi}^{(n)} = \delta\left(\vec{K} + \vec{k}_{K^+} - \vec{k}_{\pi^-}\right)\int d\vec{\tilde{q}}\left\langle \Psi_f^n \middle| \vec{\tilde{q}}' \right]\left[\vec{\tilde{q}}_1' \middle| t_{\pi^- p} \middle| \vec{\tilde{q}}_1 \right]\left[\vec{\tilde{q}} \middle| \Psi_i \right\rangle \tag{1}$$

where $\vec{\tilde{q}}_1$, $\vec{\tilde{q}}$, $\vec{\tilde{q}}'$ and $\vec{\tilde{q}}_1'$ are the relative momentum and $\left[\vec{\tilde{q}}_1' \middle| t_{\pi^- p} \middle| \vec{\tilde{q}}_1\right]$ is the two-body t-matrix for the elementary process $\pi^- + p \rightarrow K^+ + \Sigma^-$. Differential cross section of (π^-, K^+) is written as

$$d^3\sigma = \frac{(2\pi)^4}{\hbar^2 k_{\pi^-} c^2} E_{\pi^-} \left|\left\langle t_{\pi^- p}\right\rangle\right|^2 d\vec{k}_{K^+}\left(-\frac{1}{\pi}\right)\text{Im}\left[\int d\vec{r}\int d\vec{r}'\, f^*(\vec{r}')\left\langle \vec{r}' \middle| \frac{1}{E - H_{Y-Nucl} + i\varepsilon} \middle| \vec{r}\right\rangle f(\vec{r})\right] \tag{2}$$

where $f(\vec{r}) = \Psi_i(\vec{r}_i)\exp\left\{i\eta(\vec{k}_{\pi^-} - \vec{k}_{K^+})\bullet\vec{r}\right\}$,

$$f^*(\vec{r}) = \Psi_i^*(\vec{r}_i{}')\exp\left\{-i\eta(\vec{k}_{\pi^-} - \vec{k}_{K^+})\bullet\vec{r}'\right\}, \quad \eta = \frac{M_{Nucl}}{M_{Nucl} + M_Y}$$

$G^+(\vec{r}', \vec{r}) = \left\langle \vec{r}' \middle| \frac{1}{E - H_{Y-Nucl} + i\varepsilon} \middle| \vec{r}\right\rangle$ is the Green's-function. The differential cross-section of missing-mass spectrum is obained by using Green's function method as follows;

$$\frac{d^2\sigma}{dYc^2.d\cos\theta_{K^+}} = \frac{(2\pi)^5}{\hbar^2 k_{\pi^-} c^2}\frac{E_{\pi^-}\vec{k}_{K^+}\left|\left\langle t_{\pi^- p}\right\rangle\right|^2}{\left\{\frac{\hbar^2 c^2}{E_{K^+}} + \frac{\hbar^2}{M_{Y-Nucl}}\left(1 - \frac{k_{\pi^-}}{k_{K^+}}\cos\theta_{K^+}\right)\right\}} \tag{3}$$

$$\times\left(-\frac{1}{\pi}\right)\text{Im}\left[\sum_{\ell=0}^{\infty}(2\ell+1)\int dr\int dr'\, j_\ell^*(Qr')u_i^*(r')\frac{u_\ell^0(r_<)u_\ell^+(r_>)}{W(u_\ell^0, u_\ell^+)}j_\ell(Qr)u_i(r)\right].$$

Let u_ℓ^0 and u_ℓ^+ are the solutions of $\left[k^2 + \frac{d^2}{dr^2} - \frac{\ell(\ell+1)}{r^2} - \tilde{U}(\vec{r})\right]$ which satisfy the boundary condition at origin and asymptotic region respectively. The second term of the above equation is also known as the spectral function, S(E). Differential cross-section of the reaction (π^-, K^+) versus missing mass of Σ^--^{8}Li system is numerically calculated by using equation (3). By substituting the various emitted particle energy E_K into Eq. (3), we obtained the spectrum of differential cross-section versus missing-mass. In the numerical calculation, missing mass value is controlled by varying the $E_{Y\text{-}Nucl}$ from minimum value -100 MeV to maximum value 100 MeV with the interval of energy steps 2 MeV. The corresponding values of missing mass are from 8570 MeV to 8770 MeV.

^{9}Be Target

The single-particle wave function of proton for $0p_{3/2}$ ($0s_{1/2}$) state in ^{9}Be target is calculated by the Woods-Saxon potential with a= 0.65 fm, R=1.25 $A^{1/3}$ fm. The strength parameter of the potential must be adjusted to be V_0=-59.6 MeV and V_1=-30.0MeV with $V_{0p} = V_0 + \frac{(N-Z)}{A}V_1$ for the protons in the $p_{3/2}$ ($s_{1/2}$) state, and V_{so}^N = 22 MeV [1] in order to reproduce the data of proton single-particle energies in ${}^9\text{Be}(\gamma, p)\,{}^8\text{Li}$ reaction [2].We obtain the single particle energies of -21.793 MeV for $0s_{1/2}$ and -16.88 MeV for $0p_{3/2}$ and the charge radius for ${}^9\text{Be}(3/2^-)$

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becomes 2.47fm. This value in the single particle model is consistent with that of 2.5 ± 0.09 fm in electron elastic scatterings [3].

Σ-Nucleus Potential

The Σ -nuclear state is obtained by solving the Schrödinger equation

$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 + \mathrm{U}_\Sigma(\mathrm{r}) + \mathrm{V}_{\mathrm{Coul}}(\mathrm{r})\right] \Psi_\Sigma = E\ \Psi_\Sigma,$$

where μ is the Σ-nucleus reduced mass, U_Σ is the Σ-nucleus potential, and V_{Coul} is the Coulomb potential. Here the Σ-nucleus potential is given as $\mathrm{U}_\Sigma(\mathrm{r}) = [V_\Sigma + i\, W_\Sigma]\, f(r)$ with the Woods-Saxon form as $f(r) = \left\{1 + \exp\left[\frac{r-R}{a}\right]\right\}^{-1}$ where $R = r_0 A_{core}^{1/3}$ and the radius r_0=1.1fm and diffuseness of the potential, a=0.67 fm respectively. The ground state of ^{8}Li as the nuclear core is known to be a $J^\pi=2^+$ and isospin T=1 with half life $\tau_{1/2}$= 839.9 ±0.9 ms [2]. For V_{Coul}, we use an attractive Coulomb potential with the uniform distribution of a charged sphere where Z = 3 for Σ^--^{8}Li. The strength parameter of (V_Σ, W_Σ) can be adjusted appropriately to reproduce available bound states.

RESULTS

We have calculated the missing mass spectrum for Σ^--^{8}Li system by applying the Harada's Woods-Saxon potential [4] which provides the ability to explain the data of the continuum spectrum in Σ and Λ regions of J-PARC E10 experiment [5]. The strength of this potential is V_Σ=+30±10 MeV and W_Σ= -26±2 MeV with R=0.835 $A_{\mathrm{core}}^{1/3}$ fm and a=0.706 fm. There is no bound state in the spectral function of Σ^-- ^{8}Li production which is described in Figure 2. (a). The average Σ-nucleus potential must be shallow with well depth of 10≤ V≤ 25 MeV [6]. It is found that bound states appear with E = -3.2 MeV and Γ=2.58 MeV for $U_{\Sigma\text{-Nucl}}$(r)=(-20.0,-3.0) MeV and E= -0.4 MeV, and Γ=2.5 MeV for $U_{\Sigma\text{-Nucl}}$(r)= (-12.0-3.0) MeV respectively.

Σ^- is a doorways state for Λ-hypernucleus through Σ -Λ coupling effect. The coherent Λ-Σ coupling is a process in which a nucleon remains in its ground configuration after converting Λ to Σ, while giving all other nucleons an equal footing to interact with the Σ. The incoherent Λ-Σ coupling is a process in which a neutron changes to an excited level after the interaction. Therefore we have investigated the structure of p-shell hypernuclei, namely, ${}^{9}_{\Lambda}\mathrm{He}(0^+)$ ground state has been investigated by using Brueckner-Hartree-Fock method with Shinmura's Y-N interactions NSC97f(S). The above mentioned hypernucleus is considered as a system consisting of a Λ-hyperon and a core nucleus ^{8}He resonance state. Since we are interested in Λ-Σ coupling effect, Σ channel is included in our considerations. Thus we have solved the coupled channel Schrodinger equation within the frame work of Gaussian basis treatment to take into account Σ mixing probability. In our calculation, it is found that Λ-Σ coherent coupling effect is 0.19 MeV and Σ mixing probability is 0.2%.

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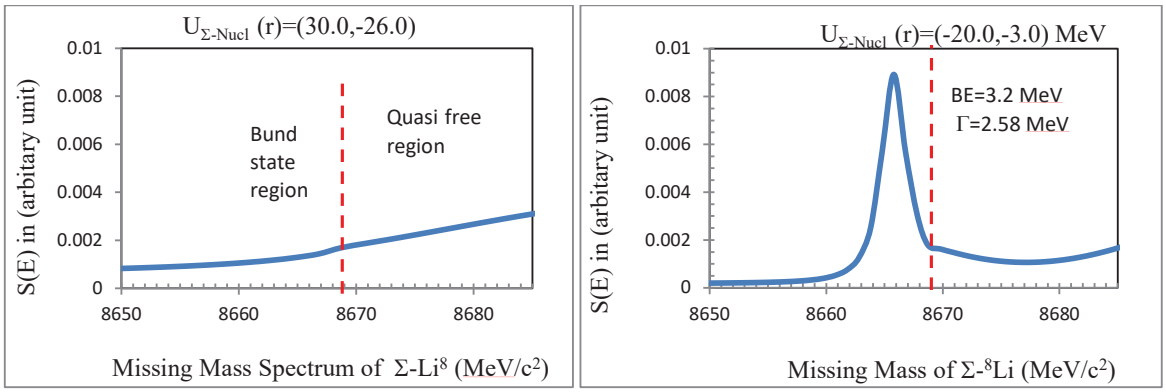

Figure 2. (a) Figure 2. (b)

Figure 2. The missing-mass spectrum for Σ^--^{8}Li (a) with repulsion and attraction of Σ-nucleus potential and (b) with attraction of Σ-nucleus potential.

CONCLUSION

We have investigated the spectral function of ^{9}Be (π^-, K$^+$) $_\Sigma{}^9$He reaction for various Σ^- - Nucleus potential strengths. We have adjusted the strength parameters of Σ^-- Nucleus potential, V_Σ and W_Σ; to find an observable narrow peak in the spectrum. When the strength of V_Σ becomes -20.0MeV and W_Σ=-3.0 MeV, a sharp peak is found at missing mass M=8666 MeV/c^2 and level width Γ=2.6 MeV respectively which corresponds to 3.2 MeV binding energy of $_\Sigma{}^9$He. By solving the coupled channel Schrodinger equation, it is found that Λ-Σ coupling effect in $_\Lambda{}^9$He is 0.19 MeV which gives the more binding in this system and Σ component is 0.2%. Σ-hyperhelium $_\Sigma{}^4$He was only observed by the (K^- stopped, π^-) reaction with binding energy $3.2 \pm 0.3^{+1.1}_{-0.1}$ MeV and level width of $4.6 \pm 0.5^{+1.6}_{-1.3}$ MeV [7]. Thus, the information of Σ^-- Nucleus potential is limited. In order to establish the strength of Σ^--Nucleus interaction, more experimental data with high statistic are awaited.

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